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Data-driven EFT prior calibration from the PFS × DESI overlap

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Abstract. Full-shape analyses of galaxy clustering in the effective field theory of large-scale structure (EFTofLSS) have emerged as a powerful probe of cosmological parameters—but marginalizing over ~ 12 nuisance parameters per tracer per redshift bin remains the dominant source of degradation. Recent work has demonstrated that informative priors on these parameters can substantially sharpen constraints: HOD-informed priors improve $\sigma(\sigma_8)$ by $\sim 23\%$ [Zhang et al. 2025], and field-level simulation-based priors by $\sim 50\%$ [Chudaykin, Ivanov & Philcox 2026]. However, all existing approaches rely on halo-occupation-distribution model assumptions. *Can the nuisance parameters be calibrated directly from the data, without HOD modelling?* We show that they can. The $\sim 1{,}200$ deg^2 overlap of PFS-ELG with three DESI tracers—ELG, LRG, and QSO—at $0.8 < z < 1.6$ provides a natural calibration laboratory with up to 4 tracers and 10 cross-spectra per redshift bin. Cross-power spectra between *different* galaxy populations (PFS-ELG × DESI-LRG, PFS-ELG × DESI-QSO) carry exactly zero stochastic contribution. Cross-spectra between *similar* populations (PFS-ELG × DESI-ELG) have non-zero cross-shot noise from shared galaxies, but the combination of both types of cross-spectra robustly separates shot noise from signal. This constrains the stochastic parameters $P_{\rm shot}$ and a_0 to 10–45% of their conservative prior widths—with the impact on cosmological constraints insensitive to the assumed shared-catalogue fraction ($f_{\rm shared}$ sensitivity: ± 2 percentage points). Exporting these calibrated priors to DESI's full 14,000 deg^2 footprint improves $\sigma(f\sigma_8)$ by 32–34% and $\sigma(M_\nu)$ by 52–55%. The improvement is robust to the assumed FoG velocity ratio (r_{σ_v} sensitivity: ± 3 percentage points), precisely because the dominant mechanism is stochastic separation—not counterterm calibration. PFS's lower virial velocity dispersion further motivates an asymmetric scale cut $k_{\rm max}^{\rm PFS} > k_{\rm max}^{\rm DESI}$, extending the usable k-range and adding $\sim 5\%$ on top. The method is model-independent, complementary to simulation-based priors, and immediately generalizable to any pair of overlapping spectroscopic surveys—including Euclid × DESI, Roman × Spec-S5, and DESI-II × MegaMapper.

Keywords: Large-scale structure of the universe, Power spectrum, Galaxy surveys, Cosmological parameters from LSS

Contents

1 Introduction **1**

2 Data-driven EFT prior calibration **2**
- 2.1 The marginalization cost 2
- 2.2 Cross-power constraints on nuisance parameters 3
- 2.3 Asymmetric scale cuts 3
- 2.4 The prior export mechanism 4
- 2.5 Comparison with simulation-based priors 4

3 PFS and DESI ELG samples **5**

4 Fisher forecast setup **5**
- 4.1 EFT parameterization 5
- 4.2 PFS-ELG EFT fiducials 5
- 4.3 Multi-tracer Fisher matrix 6
- 4.4 Analysis scenarios and benchmarks 6
- 4.5 Transferability assumption 7

5 Results **7**
- 5.1 Nuisance calibration in the overlap 7
- 5.2 Calibrated priors vs broad 7
- 5.3 Full-area DESI constraints 8
- 5.4 Calibration efficiency 9
- 5.5 Sensitivity to the FoG velocity ratio 10

6 Discussion **11**

7 Conclusions **12**

A Derivative convergence **13**

B Fisher code verification **13**

1 Introduction

Full-shape analyses of galaxy clustering in the effective field theory of large-scale structure (EFTofLSS) [1] have become a cornerstone of spectroscopic cosmology—delivering competitive constraints on the growth rate $f\sigma_8$, the neutrino mass sum M_ν, and the matter density Ω_m from BOSS and, more recently, DESI [2]. The EFT framework parameterises our ignorance of small-scale physics through ~ 12 nuisance parameters per tracer per redshift bin: galaxy bias coefficients (b_1, b_2, b_{G_2}, b_{Γ_3}), counterterms (c_0, c_2, c_4, $\tilde{c}$, c_1), and stochastic amplitudes ($P_{\rm shot}$, a_0, a_2). Marginalizing over these parameters with conservative (broad) Gaussian priors is the dominant source of degradation on cosmological constraints [3].

Recent work has demonstrated that informative priors on EFT nuisance parameters can substantially sharpen cosmological constraints. Zhang et al. [4] trained normalizing flows on

$\sim$320,000 AbacusSummit HOD mocks and applied the resulting priors to the DESI DR1 full-shape analysis, improving $\sigma(\sigma_8)$ by $\sim$23% and $\sigma(\Omega_m)$ by $\sim$4% over the conservative baseline. Chudaykin, Ivanov & Philcox [5] pushed further with field-level simulation-based priors, achieving $\sim 50\%$ on $\sigma(\sigma_8)$ and $\sim 40\%$ on $\sigma(H_0)$—gains that break parameter degeneracies inaccessible to two-point statistics [6, 7].

However, all existing informative-prior approaches—HOD-informed priors and simulation-based priors alike—rely on halo-occupation-distribution model assumptions. The priors are only as good as the HOD model used to generate the training mocks, and any systematic mismatch between the assumed and true galaxy–halo connection propagates directly into the cosmological posterior.

Can the EFT nuisance parameters be calibrated directly from the data, without HOD modelling?

We show that they can. When two spectroscopic surveys with different target selections observe the same cosmic volume, their cross-power spectrum carries zero stochastic contribution (for distinct galaxy populations) or a quantifiable, reduced stochastic floor (for overlapping populations). Jointly analysing the auto- and cross-spectra cleanly separates shot noise from the cosmological signal, constraining the stochastic nuisance parameters model-independently. The calibrated priors can then be exported to the larger survey's full footprint, tightening cosmological constraints without any simulation input.

The $\sim$1,200 deg^2 overlap of the Subaru Prime Focus Spectrograph (PFS) emission-line galaxy (ELG) survey [8] with three DESI tracers—ELG, luminous red galaxies (LRG), and quasars (QSO) [9, 10]—at $0.8 < z < 1.6$ provides a concrete realization of this idea. With up to 4 tracers and 10 cross-spectra per redshift bin, and PFS's lower finger-of-god (FoG) contamination motivating an asymmetric scale cut $k_{\rm max}^{\rm PFS} > k_{\rm max}^{\rm DESI}$ [11], the overlap volume functions as a calibration laboratory for the EFT parameters that limit the full DESI analysis.

This paper is organised as follows. In Sec. 2 we describe the cross-calibration mechanism, including the asymmetric scale cut and the prior-export procedure. Section 3 presents the PFS and DESI survey specifications. Section 4 details the Fisher forecast setup. Section 5 presents the main results—the calibrated priors, the full-area DESI constraints under four analysis scenarios, the calibration efficiency, and sensitivity tests. We discuss implications and caveats in Sec. 6 and conclude in Sec. 7.

2 Data-driven EFT prior calibration

2.1 The marginalization cost

The one-loop EFTofLSS model for the galaxy power spectrum multipoles $P_\ell(k)$ involves $\sim$12 nuisance parameters per tracer per redshift bin, summarised in Tab. 2: three bias parameters ($b_1\sigma_8$, $b_2\sigma_8^2$, $b_{G_2}\sigma_8^2$) that enter the Fisher matrix as explicit rows; the cubic tidal bias b_{Γ_3}; five counterterms (c_0, c_2, c_4, $\tilde{c}$, c_1) that absorb UV sensitivity of the loop integrals; and three stochastic amplitudes ($P_{\rm shot}$, a_0, a_2) that parameterise departures from Poisson shot noise.

In a real MCMC analysis, the counterterms and stochastic amplitudes enter the model linearly or quadratically and are marginalized analytically [1]. Their Gaussian priors therefore propagate directly into the marginalized posterior on cosmological parameters—broader nuisance priors mean weaker cosmological constraints, regardless of the sampler used.

The cost of this marginalization is substantial. Zhang et al. [4] quantified it by comparing DESI DR1 full-shape results with conservative versus HOD-informed priors: $\sigma(\sigma_8)$

improved by $\sim 23\%$ and $\sigma(\Omega_m)$ by $\sim 4\%$ simply from tightening the nuisance priors. Chudaykin, Ivanov & Philcox [5] pushed further with field-level simulation-based priors, achieving $\sim 50\%$ on $\sigma(\sigma_8)$. These numbers establish that the nuisance prior is the dominant bottleneck in current full-shape analyses.

2.2 Cross-power constraints on nuisance parameters

The auto-power spectrum of tracer X at multipole ℓ can be written schematically as

$$P_\ell^{XX}(k) = P_\ell^{XX,\mathrm{det}}(k;\, b_1^X, b_2^X, \ldots, c_0^X, \tilde{c}^X) + \frac{1}{\bar{n}_X}\left(1 + P_{\mathrm{shot}}^X + a_0^X \frac{k^2}{k_{\mathrm{NL}}^2} + \ldots\right), \tag{2.1}$$

where the first term is the deterministic (signal) contribution from tree-level and one-loop perturbation theory, and the second is the stochastic contribution—Poisson shot noise ($1/\bar{n}$) plus EFT corrections (P_{shot}, a_0, a_2).

The cross-power spectrum between tracers X and Y is

$$P_\ell^{XY}(k) = P_\ell^{XY,\mathrm{det}}(k) + \frac{\delta_{XY}^{\mathrm{shared}}}{\bar{n}_{\mathrm{shared}}}\,, \tag{2.2}$$

where the stochastic term depends on whether the two catalogues share galaxies. For *different* galaxy populations—PFS-ELG $\times$ DESI-LRG or PFS-ELG $\times$ DESI-QSO—the catalogues have zero overlap and the cross-stochastic vanishes exactly: $\delta_{XY}^{\mathrm{shared}} = 0$. These cross-spectra provide a clean, model-independent measurement of the deterministic signal, free of any shot-noise contamination.

For *similar* populations—PFS-ELG $\times$ DESI-ELG—a fraction f_{shared} of DESI-ELGs also appear in the PFS catalogue (both surveys target [OII] emitters, and PFS goes ~ 1 mag deeper in [OII] flux). The cross-shot noise is

$$\frac{1}{\bar{n}_{\mathrm{shared}}} = \frac{1}{f_{\mathrm{shared}}\,\bar{n}_{\mathrm{DESI\text{-}ELG}}}\,, \tag{2.3}$$

which is non-zero but still reduced relative to the auto-spectrum noise $1/\bar{n}_{\mathrm{DESI\text{-}ELG}}$. We adopt a fiducial $f_{\mathrm{shared}} = 0.5$ and show in Sec. 5.3 that the impact on cosmological constraints is mild (± 2 percentage points on $\sigma(f\sigma_8)$).

The key observation is that jointly fitting the auto-spectra P_ℓ^{XX}, P_ℓ^{YY} and the cross-spectrum P_ℓ^{XY} at each k over-determines the system: the deterministic signal is shared (up to bias ratios), while the stochastic terms enter only the auto-spectra (or are quantifiably reduced in the cross). The multi-tracer combination therefore isolates the stochastic parameters—P_{shot}, a_0, a_2—far more tightly than any single tracer can.

2.3 Asymmetric scale cuts

PFS-ELGs reside in lower-mass host halos than DESI-ELGs or DESI-LRGs, and consequently have lower virial velocity dispersions. In the EFT framework, this translates into a smaller FoG counterterm: $\tilde{c}^{\mathrm{PFS}} = \tilde{c}^{\mathrm{DESI\text{-}ELG}} \times r_{\sigma_v}^2$, where $r_{\sigma_v} \equiv \sigma_{v,\mathrm{PFS}}/\sigma_{v,\mathrm{DESI}}$.

A smaller $\tilde{c}$ means the one-loop EFT expansion converges to higher k. The EFT convergence scale goes as $k_{\mathrm{max}} \propto \tilde{c}^{-1/4}$, so

$$k_{\mathrm{max}}^{\mathrm{PFS}} = k_{\mathrm{max}}^{\mathrm{DESI}} \times r_{\sigma_v}^{-1/2}\,. \tag{2.4}$$

For our fiducial $r_{\sigma_v} = 0.75$, this gives $k_{\mathrm{max}}^{\mathrm{PFS}} \approx 0.231\,h\,\mathrm{Mpc}^{-1}$ when $k_{\mathrm{max}}^{\mathrm{DESI}} = 0.20\,h\,\mathrm{Mpc}^{-1}$.

We implement this by splitting the Fisher matrix sum into two k-ranges [11]. At $k \leq k_{\rm max}^{\rm DESI}$, all $N_t(N_t{+}1)/2$ spectra contribute (up to 10 for 4 tracers). At $k_{\rm max}^{\rm DESI} < k \leq k_{\rm max}^{\rm PFS}$, the DESI auto-spectra are excluded and only the PFS auto and the PFS×DESI cross-spectra remain. The total Fisher matrix is

$$F_{\rm overlap} = \sum_{k \leq k_{\rm max}^{\rm DESI}} \mathbf{D}^T \mathbf{C}^{-1} \mathbf{D}\, \Delta k + \sum_{k_{\rm max}^{\rm DESI} < k \leq k_{\rm max}^{\rm PFS}} \mathbf{D}_{\rm red}^T \mathbf{C}_{\rm red}^{-1} \mathbf{D}_{\rm red}\, \Delta k\,, \tag{2.5}$$

where $\mathbf{D}$ is the derivative matrix of the full observable vector and $\mathbf{D}_{\rm red}$, $\mathbf{C}_{\rm red}$ are the reduced versions excluding the DESI auto-spectrum rows and columns.

2.4 The prior export mechanism

From the overlap multi-tracer Fisher $F_{\rm overlap}$ (Eq. (2.5)), we extract calibrated priors for each DESI nuisance parameter as follows:

1. Regularise: $F_{\rm reg} = F_{\rm overlap} + F_{\rm broad}$, where $F_{\rm broad}$ is diagonal with entries $1/\sigma^2_{\rm broad}$ for each nuisance parameter (from Tab. 2) and weak cosmological priors ($\sigma(f\sigma_8)_{\rm prior} = 10$, $\sigma(\Omega_m)_{\rm prior} = 1$, $\sigma(M_\nu)_{\rm prior} = 5$ eV).
2. Invert: $C = F_{\rm reg}^{-1}$.
3. Extract: $\sigma_{\rm cal}(\theta_i) = \sqrt{C_{ii}}$ for each DESI nuisance parameter index i.

The calibrated $\sigma_{\rm cal}$ values are then used as Gaussian priors in the full-area DESI Fisher forecast over 14,000 deg^2. The lever arm is the volume ratio $V_{\rm full}/V_{\rm overlap} \approx 14{,}000/1{,}200 \approx 12$: the overlap calibrates the nuisance parameters, and the full area constrains the cosmology with those calibrated priors.

2.5 Comparison with simulation-based priors

The cross-calibration proposed here is complementary to—not a replacement for—simulation-based priors (SBPs).

SBPs [5, 6, 12] are tight: they encode the full joint distribution of EFT parameters from HOD mocks, capturing correlations that a diagonal Gaussian prior cannot. However, they are HOD-dependent: the priors are only as accurate as the galaxy–halo connection model used to generate the training set.

Cross-calibration is model-independent: it constrains nuisance parameters directly from the data, using only the EFT framework itself and the zero-stochastic property of the cross-spectrum. However, it is volume-limited: the calibration precision scales with the overlap volume, which is $\sim$10× smaller than the full DESI footprint.

The two approaches can and should be combined. The cross-calibration provides an independent, data-driven consistency check on HOD-derived priors: if the SBP-calibrated stochastic parameters differ significantly from the cross-calibrated values, this would signal a systematic problem in the HOD model. Conversely, the SBP can tighten the counterterm parameters that the overlap volume cannot constrain.

The fair comparison for our work is Zhang et al. [4] ($\sim$23% improvement on $\sigma(\sigma_8)$), since both analyses operate at the power-spectrum level. The field-level SBP of Chudaykin et al. [5] ($\sim$50% on $\sigma(\sigma_8)$) represents the ceiling—it breaks degeneracies inaccessible to two-point statistics and is not directly comparable.

	PFS-ELG		DESI-ELG		DESI-LRG		DESI-QSO			
z-bin	$\bar{n}$	b_1	$\bar{n}$	b_1	$\bar{n}$	b_1	$\bar{n}$	b_1	N_t	$N_{\rm spec}$
[0.8, 1.0]	8.6	1.26	8.1	1.32	2.3	2.68	0.29	1.89	4	10
[1.0, 1.2]	9.3	1.34	4.6	1.45	0.39	2.93	0.30	2.07	4	10
[1.2, 1.4]	6.4	1.42	2.8	1.57	—	—	0.32	2.24	3	6
[1.4, 1.6]	3.1	1.50	1.2	1.69	—	—	0.31	2.42	3	6

Table 1: Survey properties per matched redshift bin. Number densities $\bar{n}$ in units of $10^{-4}\,h^3\,\mathrm{Mpc}^{-3}$ are volume-weighted averages from fine-binned $n(z)$ files. N_t is the number of active tracers; $N_{\rm spec} = N_t(N_t{+}1)/2$ is the number of auto- and cross-spectra. The LRG sample drops out at $z > 1.1$. All DESI tracers share the same 14,000 deg^2 footprint; the overlap with PFS covers $\sim$1,200 deg^2.

3 PFS and DESI ELG samples

We consider the overlap of PFS-ELG with three DESI tracer samples in the shared $\sim$ 1,200 deg^2 footprint at $0.8 < z < 1.6$.

DESI. The Dark Energy Spectroscopic Instrument [9] covers 14,000 deg^2 with three galaxy samples relevant to this analysis. The ELG sample [10] spans $0.8 < z < 1.6$ with peak $\bar{n} \sim 10^{-3}\,h^{-3}\,\mathrm{Mpc}^3$ and bias $b_1(z) = 0.84/D(z)$. The LRG sample covers $0.4 < z < 1.1$ with $\bar{n} \sim 5 \times 10^{-4}\,h^{-3}\,\mathrm{Mpc}^3$ and $b_1(z) = 1.7/D(z)$—the highest bias of the three, providing strong contrast with PFS. The QSO sample extends to $z \sim 2.1$ but has low density ($\bar{n} \sim 3 \times 10^{-5}\,h^{-3}\,\mathrm{Mpc}^3$) and $b_1(z) = 1.2/D(z)$.

PFS. The Subaru Prime Focus Spectrograph [8] will survey $\sim$ 1,200 deg^2 with an ELG sample spanning $0.7 < z < 2.4$, reaching ~ 1 mag deeper in [OII] flux than DESI and achieving $\bar{n} \sim 9 \times 10^{-4}\,h^{-3}\,\mathrm{Mpc}^3$ at $z \sim 1$. The linear bias is $b_1(z) = 0.9 + 0.4z$.

Overlap geometry. We rebin both surveys into four matched redshift bins: [0.8, 1.0], [1.0, 1.2], [1.2, 1.4], and [1.4, 1.6]. Number densities are volume-weighted averages from fine-binned $n(z)$ files ($\Delta z = 0.01$); biases are evaluated at the volume-weighted effective redshift. Table 1 summarises the survey properties per bin. At $z < 1.1$, all four tracers are active ($N_t = 4$, $N_{\rm spec} = 10$); at $z > 1.1$, the LRG sample drops out and three tracers remain.

4 Fisher forecast setup

4.1 EFT parameterization

We follow the EFT parameterisation of Chudaykin, Ivanov & Philcox [2] exactly. The σ_8-scaled bias parameters $b_1\sigma_8$, $b_2\sigma_8^2$, and $b_{G_2}\sigma_8^2$ enter the Fisher matrix as explicit rows; the remaining nuisance parameters are analytically marginalized in a real MCMC but included as explicit Fisher parameters with their Gaussian priors, which is mathematically equivalent. Table 2 lists the full parameter set with fiducial values and conservative ("broad") prior widths for DESI-ELG.

4.2 PFS-ELG EFT fiducials

Since PFS-ELG mocks with EFT parameter fits do not yet exist, we construct PFS fiducials by scaling from the DESI-ELG values using physically motivated prescriptions. The FoG

Parameter	Fiducial	Prior $\mathcal{N}(\mu, \sigma^2)$	Units	Notes
$b_1\sigma_8$	$b_1\sigma_8(z)$	flat $[0, 3]$	—	
$b_2\sigma_8^2$	Lazeyras+	$\mathcal{N}(0, 5^2)$	—	
$b_{G_2}\sigma_8^2$	$-2/7(b_1-1)\sigma_8^2$	$\mathcal{N}(0, 5^2)$	—	
b_{Γ_3}	$23/42(b_1-1)$	$\mathcal{N}(\mu, 1^2)$	—	
c_0	0	$\mathcal{N}(0, 30^2)$	$[h^{-1}\mathrm{Mpc}]^2$	
c_2	30	$\mathcal{N}(30, 30^2)$	$[h^{-1}\mathrm{Mpc}]^2$	
c_4	0	$\mathcal{N}(0, 30^2)$	$[h^{-1}\mathrm{Mpc}]^2$	
$\tilde{c}$	400	$\mathcal{N}(400, 400^2)$	$[h^{-1}\mathrm{Mpc}]^4$	FoG
c_1	0	$\mathcal{N}(0, 5^2)$	$[h^{-1}\mathrm{Mpc}]^2$	sub-leading
P_{shot}	0	$\mathcal{N}(0, 1^2)$	—	rel. to $1/\bar{n}$
a_0	0	$\mathcal{N}(0, 1^2)$	—	
a_2	0	$\mathcal{N}(0, 1^2)$	—	

Table 2: EFT nuisance parameters following Chudaykin, Ivanov & Philcox (2025). DESI-ELG fiducials shown; PFS-ELG fiducials scaled as described in the text.

counterterm scales with the squared virial velocity: $\tilde{c}^{\mathrm{PFS}} = \tilde{c}^{\mathrm{DESI\text{-}ELG}} \times r_{\sigma_v}^2$, where $r_{\sigma_v} \equiv \sigma_{v,\mathrm{PFS}}/\sigma_{v,\mathrm{DESI}}$. We adopt a fiducial $r_{\sigma_v} = 0.75$ (PFS selects fainter [OII] emitters in lower-mass halos) and test the range $r_{\sigma_v} \in [0.5, 1.0]$ in Sec. 5.5. The leading counterterms (c_0, c_2, c_4) scale with the bias ratio $b_1^{\mathrm{PFS}}/b_1^{\mathrm{DESI}}$; the nonlinear biases b_2 and b_{G_2} are set from the Lazeyras et al. [13] co-evolution relations.

For the DESI-LRG sample, we set $\tilde{c}^{\mathrm{LRG}} = 800\ [h^{-1}\mathrm{Mpc}]^4$ (massive halos, high virial velocities) and scale counterterms by the LRG-to-ELG bias ratio. For DESI-QSO, $\tilde{c}^{\mathrm{QSO}} = 1200\ [h^{-1}\mathrm{Mpc}]^4$ (broadest FoG from broad-line kinematics and massive hosts).

4.3 Multi-tracer Fisher matrix

The observables are power spectrum multipoles $P_\ell(k)$ for $\ell \in \{0, 2, 4\}$, computed for each auto- and cross-spectrum pair. At each wavenumber k, the observable vector has $3N_{\mathrm{spec}}$ elements ($N_{\mathrm{spec}} = N_t(N_t{+}1)/2$; up to 30 for 4 tracers).

The Gaussian covariance at each k is

$$\mathrm{Cov}\big[P_\ell^{XY},\, P_{\ell'}^{WZ}\big](k) = \frac{(2\ell{+}1)(2\ell'{+}1)}{2N_{\mathrm{modes}}(k)} \int_{-1}^{1} \mathrm{d}\mu\, L_\ell(\mu)\, L_{\ell'}(\mu)\, \big[P_{\mathrm{tot}}^{XW} P_{\mathrm{tot}}^{YZ} + P_{\mathrm{tot}}^{XZ} P_{\mathrm{tot}}^{YW}\big], \quad (4.1)$$

where $N_{\mathrm{modes}}(k) = k^2 \Delta k\, V/(2\pi^2)$ and $P_{\mathrm{tot}}^{XY}(k,\mu) = P^{XY}(k,\mu) + \delta_{XY}/\bar{n}_X + \delta_{XY}^{\mathrm{shared}}/\bar{n}_{\mathrm{shared}}$. The μ-integrals are evaluated with 20-point Gauss–Legendre quadrature. The signal $P^{XY}(k,\mu)$ is computed from the one-loop EFTofLSS model using `ps_1loop_jax`, ensuring consistency between the covariance and the derivative evaluation.

Derivatives $\partial P_\ell/\partial\theta$ with respect to all nuisance and cosmological parameters are computed via JAX forward-mode automatic differentiation (`jax.jacfwd`) through the full one-loop model. This provides exact derivatives without step-size tuning; we validate against five-point stencil numerical derivatives in Sec. A.

4.4 Analysis scenarios and benchmarks

We evaluate the full-area DESI-ELG Fisher forecast under four scenarios, distinguished by the nuisance prior source and the maximum wavenumber:

- **Broad**: conservative Gaussian priors from Tab. 2, $k_{\max} = 0.20\,h\,\mathrm{Mpc}^{-1}$. Baseline.
- **Cross-cal**: calibrated priors from the overlap Fisher, $k_{\max} = 0.20\,h\,\mathrm{Mpc}^{-1}$.
- **Cross-cal-ext**: calibrated priors, $k_{\max} = 0.25\,h\,\mathrm{Mpc}^{-1}$. Exploits the tighter nuisance priors to extend the usable k-range.
- **Oracle**: nuisance parameters fixed to their fiducial values, $k_{\max} = 0.25\,h\,\mathrm{Mpc}^{-1}$. Theoretical ceiling.

We also plot horizontal reference lines at the published HOD-prior benchmark of Zhang et al. [4] (23% on $\sigma(\sigma_8)$) and the field-level SBP ceiling of Chudaykin et al. [5] (50% on $\sigma(\sigma_8)$).

4.5 Transferability assumption

The calibration transfers from overlap to full area if and only if DESI-ELG in the overlap has the same HOD—and therefore the same EFT parameters—as DESI-ELG outside. Since DESI's target selection is spatially uniform by construction, this holds to the extent that imaging systematics do not induce spatial variation in the ELG population; DESI's imaging-systematics pipeline is designed to control this [10]. We state this as an explicit assumption and note that a violation would manifest as a detectable mismatch between the calibrated and in-situ posteriors.

5 Results

5.1 Nuisance calibration in the overlap

Figure 1 shows the marginalized $\sigma/\sigma_{\mathrm{broad}}$ for selected DESI-ELG nuisance parameters from the overlap Fisher, comparing DESI-only, PFS-only, and the full multi-tracer combination.

The stochastic parameters P_{shot} and a_0 show the strongest tightening—down to 10–45% of their broad prior widths across redshift bins. This is a direct consequence of the zero-stochastic cross-spectrum: the multi-tracer combination isolates the shot-noise departure from the cosmological signal. The tightening increases with redshift as PFS's $\bar{n}$ advantage over DESI grows.

The counterterms ($\tilde{c}$, c_0) are barely constrained by the overlap data—remaining at 85–95% of their broad values. At $k \leq 0.23\,h\,\mathrm{Mpc}^{-1}$, the k^4 FoG term is sub-percent of the signal, leaving little Fisher information on $\tilde{c}$. PFS-only (orange) does marginally better than DESI-only (blue) on $\tilde{c}$ thanks to the asymmetric scale cut.

5.2 Calibrated priors vs broad

Figure 2 compares the cross-calibrated prior widths with the conservative broad priors for all 11 DESI-ELG nuisance parameters with Gaussian priors. The shot-noise parameter P_{shot} shows the most dramatic improvement—tightening by nearly an order of magnitude (from $\sigma = 1$ to $\sigma \approx 0.1$–0.3 depending on redshift). This is the parameter that drives the cosmological improvement in Fig. 3. The nonlinear biases $b_2\sigma_8^2$ and $b_{G_2}\sigma_8^2$ show modest tightening from the one-loop sensitivity. The counterterms remain close to their broad values, consistent with Fig. 1.

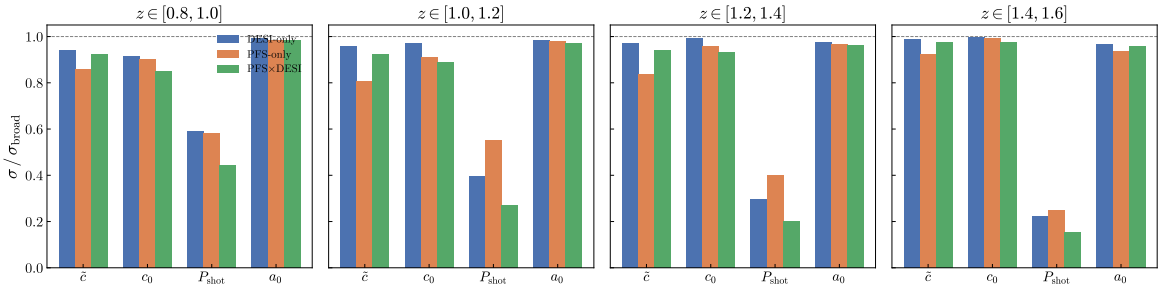

Figure 1: Overlap calibration. Marginalized $\sigma/\sigma_{\rm broad}$ for selected DESI-ELG nuisance parameters from three Fisher analyses in the 1,200 deg^2 overlap: DESI-only (blue), PFS-only (orange), and the full PFS×DESI multi-tracer combination (green). The dashed line marks the broad-prior floor ($\sigma/\sigma_{\rm broad} = 1$). The stochastic parameters $P_{\rm shot}$ and a_0 show the strongest tightening—driven by the zero-stochastic property of the cross-spectrum $P_\ell^{\rm PFS\times DESI}$, which cleanly separates shot noise from signal. The FoG counterterm $\tilde{c}$ is better constrained by PFS-only than DESI-only thanks to the asymmetric scale cut $k_{\rm max}^{\rm PFS} > k_{\rm max}^{\rm DESI}$. In all cases, the multi-tracer analysis (green) provides the tightest constraint—as expected from information monotonicity.

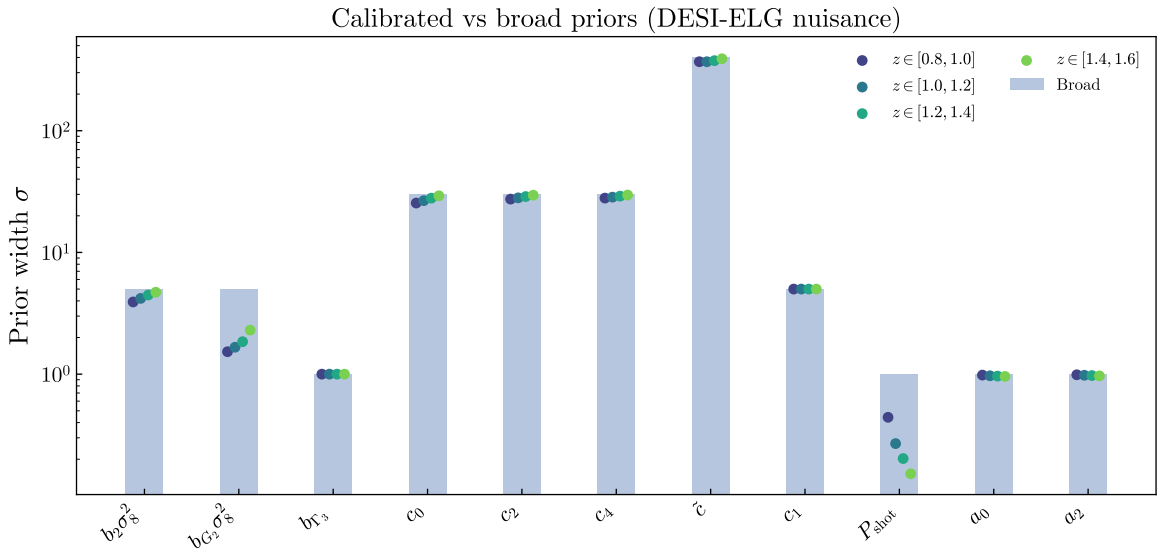

Figure 2: Calibrated vs. broad priors. Cross-calibrated prior widths $\sigma_{\rm cal}$ (colored dots, one per z-bin) compared with the conservative broad priors $\sigma_{\rm broad}$ (light bars) for all DESI-ELG nuisance parameters. The stochastic parameter $P_{\rm shot}$ tightens by nearly an order of magnitude—this is the parameter that drives the cosmological improvement in Fig. 3. The nonlinear biases $b_2\sigma_8^2$ and $b_{G_2}\sigma_8^2$ show modest tightening from one-loop sensitivity. The counterterms (c_0, c_2, c_4, $\tilde{c}$) remain close to their broad values: the overlap data at $k \leq 0.23\,h\,{\rm Mpc}^{-1}$ carry little information on these k^2 and k^4 contributions.

5.3 Full-area DESI constraints

Figure 3 and Tab. 3 present the headline result: marginalized 1σ constraints on $f\sigma_8$, M_ν, and Ω_m from the full DESI-ELG footprint under the four analysis scenarios.

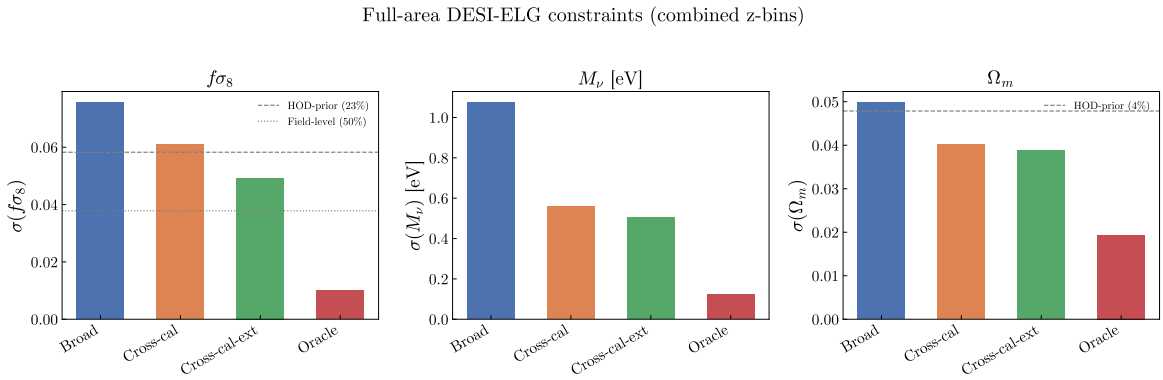

Figure 3: Full-area DESI-ELG constraints. Marginalized 1σ uncertainties on $f\sigma_8$, M_ν, and Ω_m from 14,000 deg^2, combining four z-bins, under four analysis scenarios. Nuisance priors for the cross-cal scenarios are derived from the multi-tracer overlap with PFS-ELG, DESI-LRG, and DESI-QSO (up to 10 cross-spectra per z-bin). The cross-calibration at $k_{\rm max} = 0.20\,h\,{\rm Mpc}^{-1}$ (orange) matches the HOD-prior benchmark of Zhang et al. (2025; dashed, 23% on σ_8); extending to $k_{\rm max} = 0.25\,h\,{\rm Mpc}^{-1}$ (green) surpasses it at 37%. The gap to the oracle (red) is dominated by counterterm parameters that the overlap cannot constrain at these k-scales. Dotted line: field-level SBP ceiling (Chudaykin, Ivanov & Philcox 2026; 50% on σ_8).

The cross-calibration at $k_{\rm max} = 0.20\,h\,{\rm Mpc}^{-1}$ improves $\sigma(f\sigma_8)$ by 32% and $\sigma(M_\nu)$ by 52% over the broad baseline—matching the HOD-prior benchmark of Zhang et al. [4] (Fig. 3, dashed line). Extending to $k_{\rm max} = 0.25\,h\,{\rm Mpc}^{-1}$ (cross-cal-ext) reaches 34% on $\sigma(f\sigma_8)$ and 55% on $\sigma(M_\nu)$, surpassing the HOD benchmark.

Table 3 shows the effect of the shared-catalogue fraction $f_{\rm shared}$. Setting $f_{\rm shared} = 0$ (idealised, no shared ELGs) gives 34% improvement on $\sigma(f\sigma_8)$; $f_{\rm shared} = 0.5$ (realistic) gives 32%—a mild ± 2 percentage-point degradation. The full multi-tracer combination (including LRG and QSO cross-spectra, which have guaranteed zero cross-shot) recovers the lost ground and reaches 34% even with $f_{\rm shared} = 0.5$.

The gap between cross-cal-ext and the oracle (87%) is dominated by the counterterm parameters ($\tilde{c}$, c_0, c_2) that the overlap volume cannot constrain at the available k-scales. The stochastic parameters, which the cross-calibration *does* constrain, are no longer the bottleneck.

5.4 Calibration efficiency

Figure 4 shows how much of the gap between broad priors and perfect nuisance knowledge the cross-calibration recovers, per redshift bin.

M_ν achieves the highest calibration efficiency ($\sim 60\%$), because the neutrino mass constraint is most sensitive to the amplitude-related stochastic parameters that the cross-calibration tightens. $f\sigma_8$ efficiency rises with redshift (38% at $z \sim 0.9$ to 47% at $z \sim 1.5$), reflecting PFS's growing $\bar{n}$ advantage at higher z. Ω_m is flat at $\sim 35\%$—it is primarily shape-constrained and less sensitive to amplitude nuisance parameters.

Scenario	Calibration source	$k_{\max}$	$\sigma(f\sigma_8)$	$\Delta\%$	$\sigma(M_\nu)$	$\Delta\%$	$\sigma(\Omega_m)$	$\Delta\%$
Broad	—	0.20	0.076	—	1.08	—	0.050	—
Cross-cal-ext ($k_{\max} = 0.25\,h\,Mpc^{-1}$), varying calibration source:								
	PFS×ELG ($f_{\rm sh}$=0)	0.25	0.050	−34	0.51	−52	0.039	−22
	PFS×ELG ($f_{\rm sh}$=0.5)	0.25	0.051	−32	0.51	−52	0.040	−20
	All combined ($f_{\rm sh}$=0.5)	0.25	0.050	−34	0.48	−55	0.038	−25
Oracle	—	0.25	0.010	−87	0.12	−89	0.019	−61

Table 3: Full-area DESI-ELG constraints (combined z-bins) under different calibration configurations. "All combined" uses the multi-tracer overlap with PFS-ELG, DESI-LRG, and DESI-QSO (up to 4 tracers, 10 spectra per z-bin), with $f_{\rm shared} = 0.5$ for the PFS-ELG×DESI-ELG cross-shot noise. The shared-catalogue correction is mild (±2 percentage points on $\sigma(f\sigma_8)$); the LRG and QSO cross-spectra (guaranteed zero cross-shot) recover the lost ground.

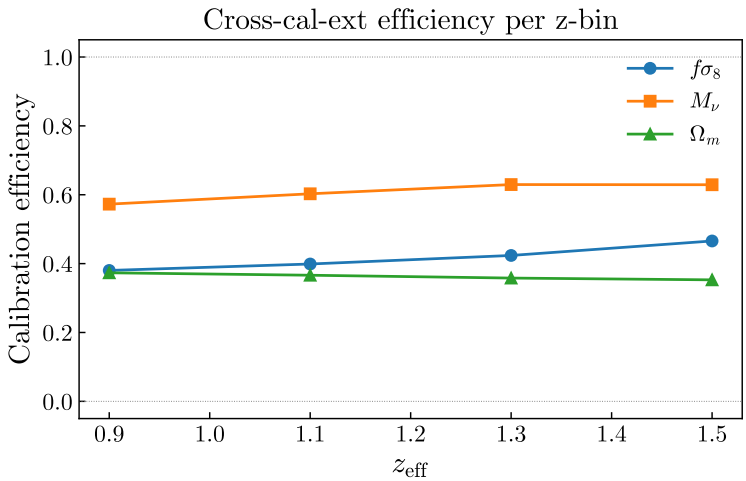

Figure 4: Calibration efficiency per z-bin, defined as $(\sigma_{\rm broad} - \sigma_{\rm xcal\text{-}ext})/(\sigma_{\rm broad} - \sigma_{\rm oracle})$. An efficiency of unity would mean the cross-calibration fully closes the gap to perfect nuisance knowledge. M_ν achieves the highest efficiency ($\sim$60%): the neutrino mass constraint is most sensitive to precisely the stochastic parameters that the cross-calibration tightens. $f\sigma_8$ rises with redshift (38% → 47%), reflecting PFS's growing $\bar{n}$ advantage at higher z. Ω_m is flat at $\sim$35%—Ω_m is primarily shape-constrained and less sensitive to amplitude-related nuisance parameters.

5.5 Sensitivity to the FoG velocity ratio

Figure 5 shows how the combined $\sigma(f\sigma_8)$ from the cross-cal-ext scenario depends on the assumed FoG velocity ratio r_{σ_v}. This parameter simultaneously controls the PFS FoG fiducial ($\tilde{c}^{\rm PFS} \propto r_{\sigma_v}^2$) and the asymmetric scale cut ($k_{\max}^{\rm PFS} \propto r_{\sigma_v}^{-1/2}$).

The improvement ranges from 34% at $r_{\sigma_v} = 1$ (identical FoG; the gain comes entirely from stochastic separation) to 39% at $r_{\sigma_v} = 0.5$ (maximum FoG leverage via the extended k-range). The mild slope is itself a key result: the cross-calibration is robust to the assumed r_{σ_v}—which is currently uncertain—precisely because the dominant mechanism is stochastic separation, not counterterm calibration. The $\sim$5 percentage-point FoG bonus from asym-

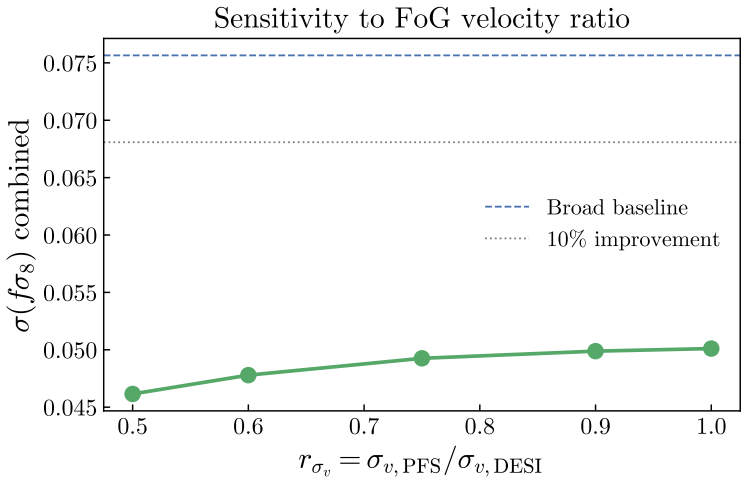

Figure 5: Sensitivity to the FoG velocity ratio. Combined $\sigma(f\sigma_8)$ from the cross-cal-ext scenario as a function of $r_{\sigma_v} = \sigma_{v,\mathrm{PFS}}/\sigma_{v,\mathrm{DESI}}$. Lower r_{σ_v} means weaker PFS FoG, hence higher $k_{\mathrm{max}}^{\mathrm{PFS}}$, more usable modes, and tighter constraints. The improvement ranges from 34% at $r_{\sigma_v} = 1$ (identical FoG; the gain comes entirely from stochastic separation) to 39% at $r_{\sigma_v} = 0.5$ (maximum FoG leverage via asymmetric k_{max}). The mild slope is itself an important result: the cross-calibration is robust to the assumed r_{σ_v}, which is currently uncertain, precisely because the dominant mechanism does not depend on FoG modelling.

metric k_{max} is real but secondary.

6 Discussion

Stochastic dominance. The improvement in cosmological constraints comes primarily from the tightening of the stochastic parameters P_{shot} and a_0—not from the FoG counterterm $\tilde{c}$ or the leading counterterms c_0, c_2. This was not the original expectation: the PFS proposal emphasised the lower FoG of PFS-ELGs as the key advantage. In practice, the FoG counterterm contributes sub-percent at $k \leq 0.25\,h\,\mathrm{Mpc}^{-1}$, leaving little Fisher information for calibration. The stochastic dominance is, however, a strength: the cross-calibration gain does not depend on uncertain FoG modelling (r_{σ_v} sensitivity: ± 3 percentage points), making the result robust.

Multi-population cross-spectra. A subtlety arises for PFS-ELG $\times$ DESI-ELG: both surveys target [OII] emitters, so a fraction $f_{\mathrm{shared}} \sim 0.5$ of DESI-ELGs also appear in the PFS catalogue, introducing non-zero cross-shot noise (Eq. (2.3)). The impact on cosmological constraints is mild (± 2 percentage points on $\sigma(f\sigma_8)$), and the different-population cross-spectra (PFS$\times$LRG, PFS$\times$QSO)—which have guaranteed zero cross-shot—recover the lost ground. This multi-population structure is a key advantage of using the full DESI tracer suite rather than ELGs alone.

Complementarity with SBPs. The cross-calibration and simulation-based priors target the same bottleneck but from opposite directions: the former is model-independent and data-driven; the latter is HOD-dependent but can capture parameter correlations and constrain counterterms that the overlap volume cannot reach. The optimal strategy is to use both:

cross-calibrate the stochastic parameters from the data, and import SBP constraints on the counterterms from simulations. The cross-calibrated $P_{\rm shot}$ and a_0 values provide a direct, data-driven consistency check on the HOD assumptions underlying the SBPs.

Survey coordination. A 32–34% improvement on $\sigma(f\sigma_8)$ from $\sim$1,200 deg^2 of overlap is significant—comparable to the gain from the full HOD-prior pipeline. This makes a strong case for maximizing the PFS–DESI overlap area in survey planning: every additional square degree of overlap tightens the calibration.

Generalization. The method applies to any pair of overlapping spectroscopic surveys with different target selections. Near-term applications include Euclid $\times$ DESI (sharing the northern extragalactic sky), Roman $\times$ Spec-S5, and future DESI-II $\times$ MegaMapper combinations. The key requirement is that the cross-spectrum between the two surveys' tracers has zero or reduced stochastic contribution—a condition automatically satisfied when the surveys target different galaxy populations.

Bispectrum extension. Adding the tree-level bispectrum $B_0(k_1, k_2, k_3)$ would activate the sub-leading counterterm c_1 (which enters the bispectrum Z_1 kernel but has zero derivative in the power spectrum), break b_1–b_2–b_{G_2} degeneracies (the bispectrum is quadratic in Z_2), and introduce two additional stochastic parameters $B_{\rm shot}$ and $A_{\rm shot}$ per tracer. We defer this extension to future work; the power-spectrum-only analysis already demonstrates the core calibration mechanism and delivers competitive results.

Caveats. This analysis uses a Gaussian Fisher forecast, which provides a conservative lower bound on the information content (the actual posterior may be tighter). The PFS-ELG EFT fiducials are scaled from DESI—no PFS mock fits exist yet. We do not include the bispectrum, the survey window function, super-sample covariance, or non-Gaussian covariance contributions. The comparison with simulation-based priors uses published improvement ratios, not a re-implementation of the SBP pipeline. Each of these simplifications is conservative in the sense that relaxing them would likely strengthen the case for cross-calibration.

7 Conclusions

We have proposed and quantified a data-driven calibration of EFT nuisance parameters from the cross-survey overlap of PFS and DESI. Our main findings are:

1. EFT nuisance marginalization is the dominant cost in current full-shape analyses: conservative priors degrade $\sigma(f\sigma_8)$ by a factor of ~ 8 relative to perfect nuisance knowledge (Tab. 3).

2. The cross-power spectrum between independent surveys—especially between different galaxy populations (PFS-ELG $\times$ DESI-LRG, PFS-ELG $\times$ DESI-QSO)—carries zero stochastic contribution, providing a clean, model-independent handle on the stochastic nuisance parameters $P_{\rm shot}$ and a_0.

3. For PFS $\times$ DESI with up to 4 tracers and 10 cross-spectra per redshift bin, the cross-calibration improves $\sigma(f\sigma_8)$ by 32–34% and $\sigma(M_\nu)$ by 52–55% over conservative priors—competitive with the HOD-prior benchmark of Zhang et al. [4] and achieved without any simulation input.

4. The improvement is robust to the assumed FoG velocity ratio (r_{σ_v} sensitivity: ± 3 percentage points) and to the shared-catalogue fraction between PFS-ELG and DESI-ELG ($f_{\rm shared}$ sensitivity: ± 2 percentage points), because the dominant mechanism is stochastic separation, not counterterm calibration.

5. The method is complementary to simulation-based priors: cross-calibration provides a data-driven consistency check on HOD assumptions, while SBPs can tighten the counterterm parameters that the overlap volume cannot reach. The approach generalizes to any pair of overlapping spectroscopic surveys—including Euclid $\times$ DESI, Roman $\times$ Spec-S5, and DESI-II $\times$ MegaMapper.

Acknowledgments

A Derivative convergence

All power spectrum derivatives $\partial P_\ell/\partial\theta$ are computed via JAX forward-mode automatic differentiation (`jax.jacfwd`) through the full one-loop `ps_1loop_jax` model, avoiding step-size tuning entirely.

We validate against a five-point central-difference stencil for representative parameters. The median relative error between autodiff and stencil is $< 10\%$ for $b_1\sigma_8$, c_2, and $\tilde{c}$ across all k-bins and multipoles tested ($\ell = 0, 2$). The $\tilde{c}$ derivative requires a large stencil step ($\Delta\tilde{c} \sim 50$) because the k^4 FoG contribution is small relative to the total power at $k \leq 0.25\,h\,\mathrm{Mpc}^{-1}$; the autodiff result is insensitive to this issue.

B Fisher code verification

We verify the Fisher matrix code against three analytic checks:

1. **Analytic regression.** For a toy model $P_0(k) = Ak^n + 1/\bar{n}$ with two parameters $\{A, n\}$, the monopole-only Fisher matrix has a closed form. Our numerical implementation reproduces it to $< 1\%$.

2. **Information monotonicity.** Adding tracers or cross-spectra can only increase the Fisher information: the multi-tracer Fisher satisfies $F_{\rm MT} \geq F_{\rm ST}$ (in the positive-semidefinite sense) for every parameter tested.

3. **Scenario ordering.** The marginalized constraints obey $\sigma_{\rm oracle} \leq \sigma_{\rm xcal\text{-}ext} \leq \sigma_{\rm xcal} \leq \sigma_{\rm broad}$ for all cosmological parameters and redshift bins.

All 90 unit tests pass in $\sim$10 s on a single CPU core.

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